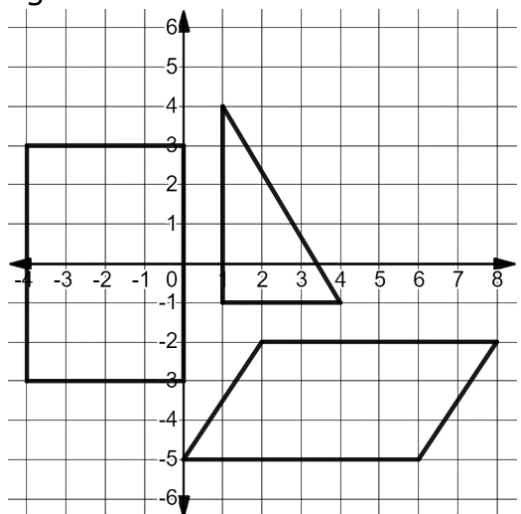


Geometry
Unit 9
Lesson 1 Homework

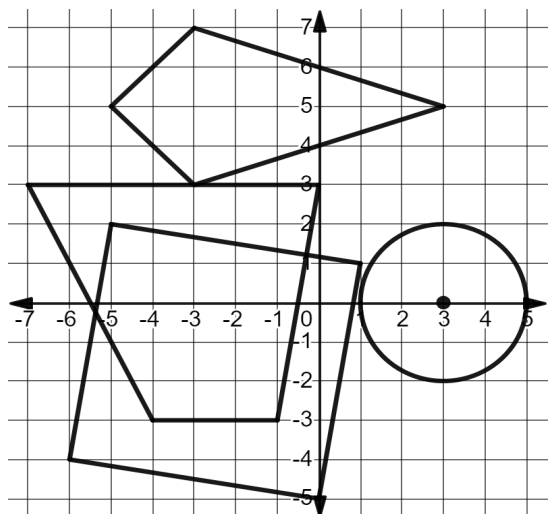
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Date _____
Period _____

A graph is shown with a rectangle, triangle, and parallelogram. Find the area of each figure.



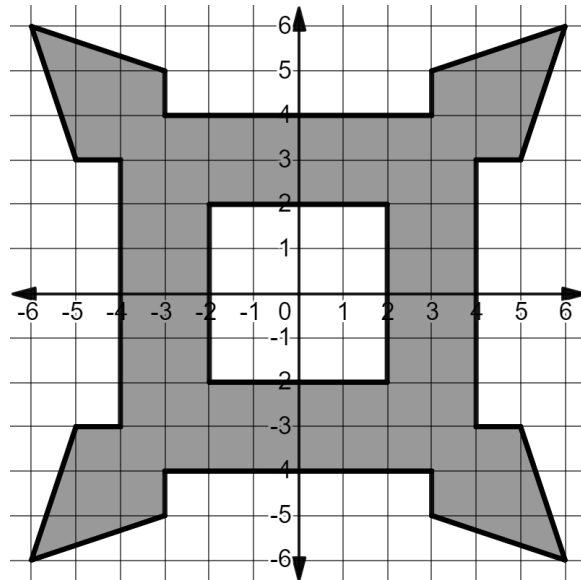
Shape	Area
Rectangle	1.
Triangle	2.
Parallelogram	3.

A graph with a trapezoid, kite, circle, and rhombus is shown. Find the area of each figure.



Shape	Area
Kite	4.
Trapezoid	5.
Rhombus	6.
Circle	7.

A graph similar to the outline of the Castillo de San Marcos in St. Augustine, Florida is shown.



8. What shapes can be used to find the area of the shaded region?
9. Divide the shaded area into smaller shapes on the graph.
10. What is the area of the shaded region?

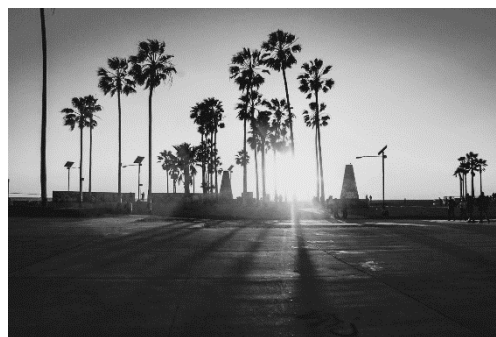
The table includes the populations of a few cities in Florida.

City	Population in 2014	Population in 2019
Destin	12,840	13,702
Panama City	36,405	36,640
Pensacola	52,505	52,642

1. Pensacola is spread over an area of 41.12 square miles. What was the population density in 2014?
2. What was the population density in 2019?
3. How did the population density change over the course of 5 years?
4. Destin is spread over an area of 7.66 square miles and Panama City covers 41.27 square miles. What was the population density of Destin in 2019?
5. What was the population density of Panama City in 2019?
6. How does the population density of Destin compare with the population density of Panama City?



7. Manatee Springs is one natural spring that provides warm water for manatees to survive. The spring run that allows water to flow from the spring into the Suwannee River is $\frac{1}{4}$ mile long. There are about 20 manatees that can be seen regularly here. If the spring run is $\frac{1}{4}$ mile wide, what is the population density of the manatees?
8. Miami is a popular tourist spot due to its beaches and warm weather. It spreads across an area of 55.25 square miles. If 24.2 million people visit the area, what is the density of visitors?
9. If only $\frac{1}{3}$ of those visitors are in Miami during the winter months, what is the population density of visitors?
10. If $\frac{3}{4}$ of the visitors are in Miami during the summer months, what is the population density of visitors?



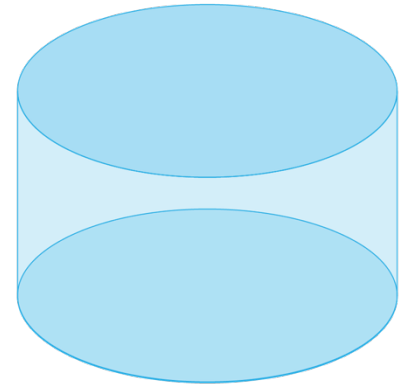
Geometry
Unit 9
Lesson 3 Homework

Name _____
Date _____
Period _____

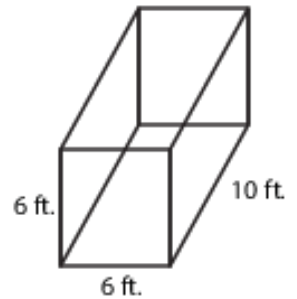
A cylinder is shown.

1. Draw a cross section parallel to the base.

2. Draw a cross section perpendicular to the base.



3. A square prism is shown with base edge length of 6 feet (ft.) and a height of 10 ft. Draw the cross section formed by the intersection of the prism and a plane parallel to the base. Label the units.



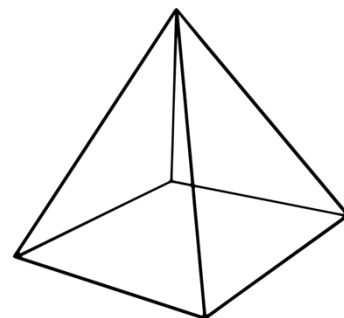
4. Draw a cross section formed by the intersection of a hexagonal prism and a plane that is parallel to the base.

5. Draw a cross section formed by the intersection of a sphere with a radius of 6 units and a plane passing through the center of the sphere. Label the units.

6. Draw a cross section formed by the intersection of a triangular prism and a plane perpendicular to its the base.

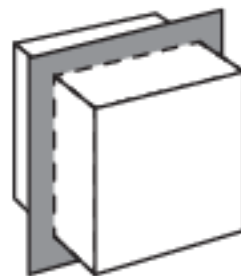
A square pyramid is shown.

7. Describe the shape of a cross section parallel to the base.

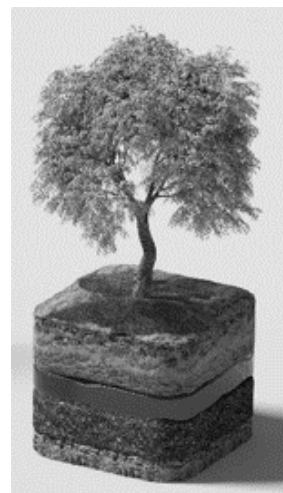


8. Describe the shape of a cross section perpendicular to the base.

9. A cube is shown, with an edge length of 10 centimeters (cm). Draw the cross section formed by the intersection of the cube and a vertical plane passing through the center of the cube.



10. What two-dimensional shape is created by a horizontal cross section going through the second layer of earth in the picture?

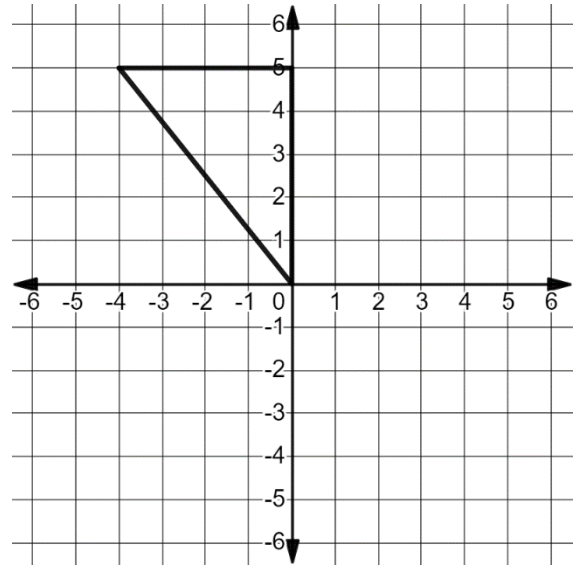


Geometry
Unit 9
Lesson 4 Homework

Name _____
Date _____
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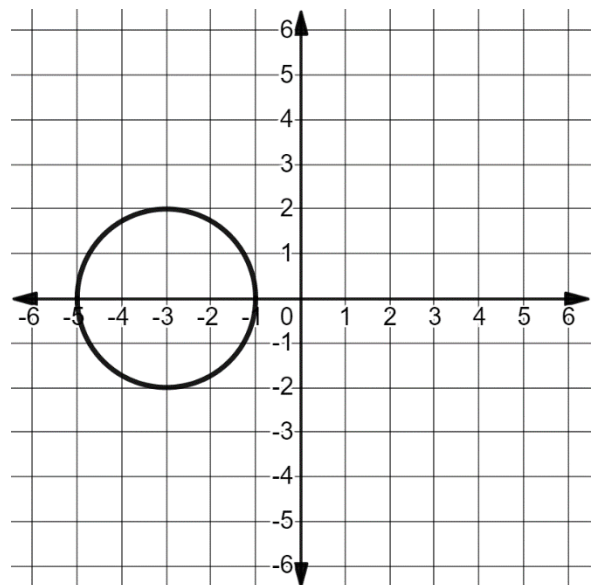
Given the graph.

1. Draw the transformation of the triangle reflected across the y -axis.
2. What three-dimensional shape would be created by rotating the triangle around the y -axis?
3. What are the base and height measurements of the three-dimensional shape, in units?
4. Draw the transformation of the triangle reflected across the x -axis in a different color.
5. What three-dimensional shape would be created by rotating the triangle around the x -axis?

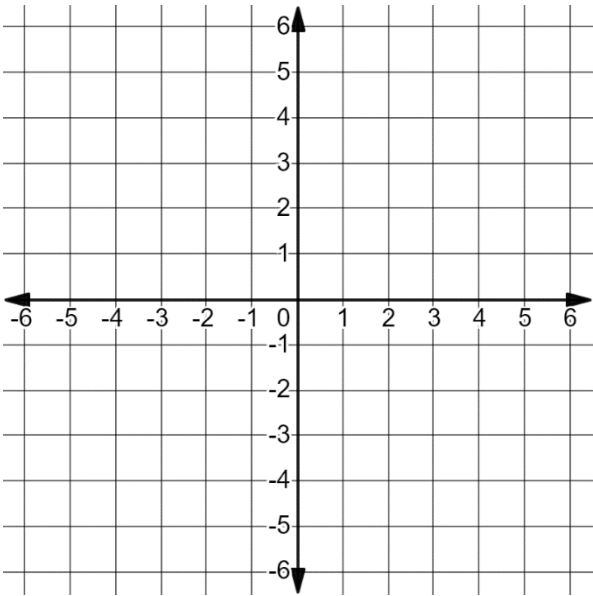
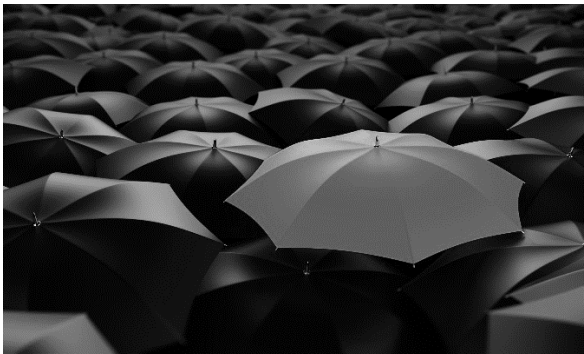
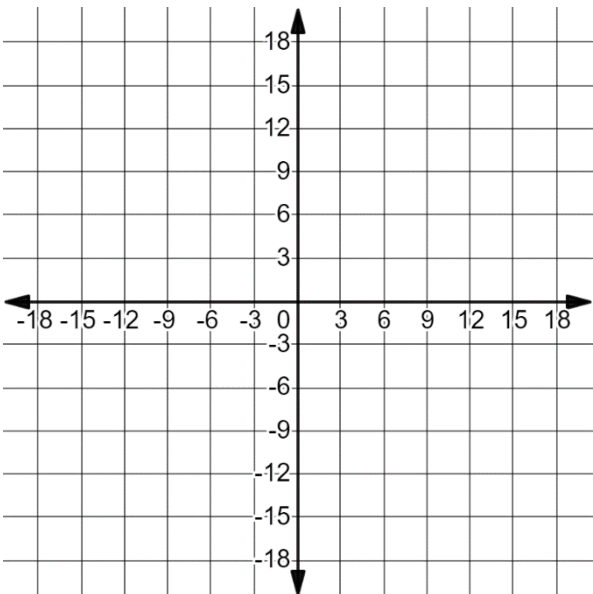


Given the circle.

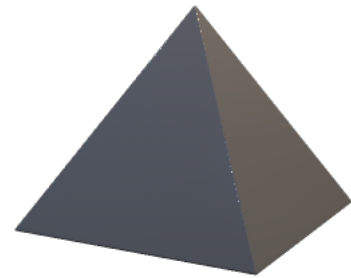
6. Draw the transformation of the circle reflected across the y -axis.
7. What three-dimensional shape would be created by rotating the circle around the y -axis?
8. Name one real world object that exhibits the three-dimensional shape created.



9. Draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

Image	Two-Dimensional Figure
A soccer ball with a radius of 4.5 inches 	
An umbrella with a radius of 18 inches 	

Using the given three figures, answer questions 1-3.



1. Which two figures can have the same volume?
2. Explain how the figures listed in Question 1 can have the same volume.
3. What information is required for each figure to determine if the volumes are equal?

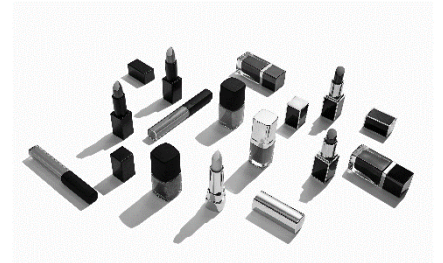
4. Isaac and Dallas went to an amusement park. They each bought a cheeseburger. Isaac's meat patty came with a square base. Dallas's meat patty had a circular base. Is it possible that they both purchased the same size of cheeseburger?



5. Dallas's meat patty was 0.75 inches high with a radius of 4.5 inches. What is the volume?

6. Isaac's hamburger patty was also 0.75 inches high but with a base whose side length was 4.5 inches. Determine if the two containers have the same volume.

7. Brand A nail polish is sold in rectangular prism jars while Brand B nail polish is sold in cylinder jars. Is it possible for a given quantity of Brand A nail polish and Brand B nail polish to have the same volume?



8. A jar of Brand B nail polish is 3 inches tall, with a diameter of 1.25 inches. In a store display, there are a total 18 jars. What is the volume of nail polish in the display?
9. One jar of Brand A nail polish is 3 inches tall, with a width of 1 inch and a length of 1.25 inches. There are 18 jars in the display. What is the total volume of nail polish in the display?
10. Compare the answers from Questions 8 and 9.

1. Find the volume of a rectangular prism with a width of 7.2 feet, a length of 6.9 feet, and a height of 10.5 feet.



2. Find the volume of a pyramid whose height is 10.5 feet and whose base is a rectangle with dimensions of 7.2 feet by 6.9 feet.

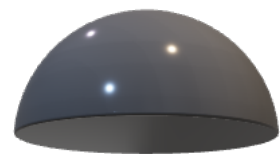


3. What is the ratio of the answer from Question 1 to the answer from Question 2?

4. Find the volume of a sphere with a diameter of 1.74 meters. Use 3.14 for π . Round to the nearest hundredth.



5. Find the volume of a hemisphere with a diameter of 1.74 meters. Use 3.14 for π . Round to the nearest hundredth.



6. What is the relationship between the answers from Questions 4 and 5?

7. Determine the volume of a cone with a radius of 16.2 inches, a height of 5.75 inches, and a slant height of 9.9 inches in terms of π .



8. Find the volume of a cylinder with a radius of 16.2 inches and a height of 5.75 inches. Round your answer to the nearest thousandth.

9. A composite figure consists of a cylinder and a square pyramid. The height of the cylinder is 58.1 centimeters. The height of the pyramid is 23 centimeters. The base of the square pyramid has a width of 19 centimeters. The base of the pyramid sits on one base of the cylinder with the width of the square base equal to the diameter of the cylinder. Find the volume of the composite figure. Round to the nearest hundredth.



10. A composite figure consists of a cylinder and a hemisphere. The height of the cylinder is 32.5 inches. The base of the hemisphere sits on one base of the cylinder, both having a diameter of 14 inches. Find the volume of the composite figure. Use 3.14 for π . Round to the nearest hundredth.



1. Oliver lives by the highway. He noticed several tractor trailers drive by every afternoon. Many of the tractor trailers are tankers. The cylinder on a tanker can be 42.6 feet long with a radius of 8.3 feet. What is the maximum volume of the cylinder on a tanker?



2. Oliver counted 16 tanker trucks one Saturday. If every tanker truck was full, what was the total volume of cargo that passed by that day?

3. A baseball has a diameter of 7.5 centimeters. What is the volume of a baseball? Round to the nearest hundredth.



4. The diameter of a softball is 8.89 centimeters. What is the volume of a softball? Round to the nearest hundredth.

5. What is the difference in volume between a baseball and a softball?

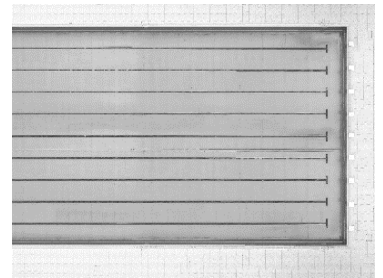
6. Penelope went on a road trip. During the drive, she noticed that some hay bales were rectangular prisms and some were cylindrical. She learned that the first hay bale she saw that was a rectangular prism was 16 inches high by 22 inches wide by 44 inches long. What was the volume of the first hay bale? Round to the nearest hundredth.

7. The second hay bale was cylindrical with a 5 foot diameter and a 4 foot width. What was the volume of the second hay bale, in cubic inches? Round to the nearest hundredth.



8. Which hay bale is larger?

9. The high school swimming pool is 25 meters long, 15 meters wide, and 3 meters deep. What is the volume of water in a full pool?



10. If the pool loses $\frac{1}{10}$ of its water depth during a butterfly swim event, what is the new volume of water?

For questions 1-10, use $\pi = 3.14$ and round all answers to the nearest thousandth if necessary.

1. Find the surface area of a pyramid with a height of 9.8 meters, slant height of 10.34 meters, and a square base with a width of 6.7 meters.



2. Find the lateral area of a pyramid whose height is 9.8 meters, slant height is 10.34 meters, and whose base is a square with a width of 6.7 meters.

3. Find the lateral area of a rectangular prism with a width of 31.8 feet, a length of 57.1 feet, and a height of 24.1 feet.



4. Find the surface area of a rectangular prism with a width of 31.8 feet, a length of 57.1 feet, and a height of 24.1 feet.

5. Find the surface area of a sphere with a diameter of 26.8 centimeters.

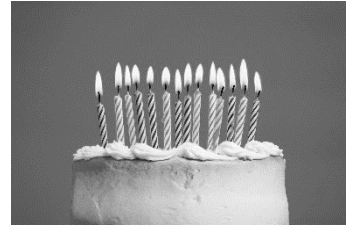


6. Find the surface area of a hemisphere with a diameter of 102.9 millimeters.

7. The surface area of a sphere is 484π inches. What is the diameter of the sphere?
8. Find the surface area of a cone with a radius of 16.7 yards, a height of 9.14 yards, and a slant height of 19.04 yards.
9. Find the lateral area of a cylinder with a radius of 6.7 inches and a height of 9.4 inches.
10. Find the surface area of a cylinder with a radius of 6.7 inches and a height of 9.4 inches.



1. Brea enjoys baking and decorating cakes. The easiest cake pan creates a cake that is 9 inches in diameter and $4\frac{1}{4}$ inches tall. What is the surface area of the cake? Round to the nearest hundredth.



2. Brea ices every possible space of the cake. She only adds her creative designs to the top of the cake. What is the area she has for her creative designs? Round to the nearest hundredth.

3. A baseball has a diameter of 7.5 centimeters. What is the surface area of a baseball? Round to the nearest hundredth.



4. The diameter of a softball is 8.89 centimeters. What is the surface area of a softball? Round to the nearest hundredth.

5. What is the difference in surface area between a baseball and a softball?

6. Newman's fence needs a new coat of paint. The fence runs along the east side of the property line which is 120 feet long. The height of the fence is 15 feet with a width of 0.5 feet. Both sides of the fence and the top need to be painted. Find the area of the fence that needs to be painted.



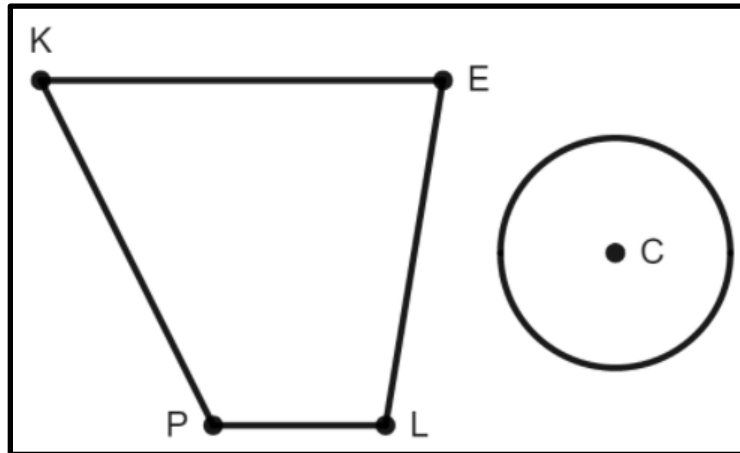
7. If 1 gallon of paint covers 410 square feet, how much paint does Newman need to buy?
8. If 1 gallon of paint costs \$23.20, how much will the paint cost?

9. Gertrude wants to build a new pen for her cow. She has one area of her property available for a rectangular pen that would measure 18 feet by 36 feet. She has a second area of her property available for a square pen that would measure 25 feet wide. Which pen area would require the least amount of fence?



10. Gertrude decides to build the pen area with the least amount of fence. If her fence is 7.5 feet tall, what is the surface area of her fence?

A diagram with Trapezoid $KELP$ and Circle C is shown. Given the dimensions, find the area of each figure and complete the table. Round to the nearest hundredth if necessary.



Shape	Dimensions	Area
Trapezoid $KELP$	$h = 52.6$, $b_1 = 47.8$, $b_2 = 39.1$	1.
Circle C	$r = 26.7$	2.

Determine the area of the dilated figures and complete the table.

Shape	Area with Scale Factor of $\frac{7}{3}$	Area with Scale Factor of $\frac{3}{7}$
Trapezoid $K'E'L'P'$	3.	4.
Circle C'	5.	6.

7. Parallelogram $CALM$ has a base of 7.6 feet and a height of 21.3 feet. What is the area of Parallelogram $CALM$?

8. Parallelogram $CALM$ is multiplied by a scale factor to create Parallelogram $C'A'L'M'$. Parallelogram $C'A'L'M'$ has an area of 485.64 square feet. What is the scale factor for the area of the pre-image to get the image?

9. Using the scale factor, what is the base of the image?

10. Using the scale factor, what is the height of the image?

For questions 1-10, use $\pi = 3.14$ and round all answers to the nearest hundredth if necessary.

1. When Felix was young, he had a mini trampoline. It was 36 inches in diameter. What was the base (jumping) area of the mini trampoline?

2. As Felix grew bigger, he needed a bigger trampoline. Felix got a new trampoline for his birthday. It was 128 inches in diameter. What was the base (jumping) area of the new trampoline?

3. What is the difference in areas of the two trampolines?

4. With the frame, Felix's new trampoline stood 35 inches high. Felix wants to put a skirt around the trampoline to prevent his pets from going underneath it while he is jumping. What is the lateral surface area of the new trampoline that is to be covered by the skirt?

5. Although his pets could not fit under the mini trampoline, what was the lateral surface area of the mini trampoline if it stood 13 inches high?

6. Hermione is decorating for a birthday party. She has four sizes of gift boxes that are to be placed in one corner, simply for decoration. The base of one is 8 inches by 12 inches, the second is 12 inches by 16 inches, the third is 18 inches by 24 inches, and the fourth is 20 inches by 20 inches. What is the base area of the smallest box?



7. What is the base area of the second box?
8. What is the base area of the third box?
9. What is the base area of the largest box?
10. Hermione designated space in the corner for the gift boxes that is 50 inches by 60 inches. Can she fit all 4 sizes of gift boxes in the designated space?